The full interval for the reflected click rate

Research direction and responsibility · Kirtus Justus. Mathematical and computational assistance · ChatGPT (OpenAI). November 7, 2026 · Port Coquitlam, British Columbia, Canada.

> With match population s=0.409375, reference response h=0.99 and ordinary click probability q=0.3631681140623368…, the exact range after the joint phase reversal is [0.449704036935…, 0.551029468634…]. All values in this interval are attainable within the stated quantum model.

Fix the reported history 111, a rank-one match projector P, a normalized conditional input ρ, and a common binary effect $0 \le E \le I$. Write $s=\mathrm{Tr}(P\rho)$, $h=\mathrm{Tr}(EP)$, $q=\mathrm{Tr}(E\rho)$ and $r=\mathrm{Tr}(EU\rho U)$, where $U=2P-I$. The earlier lower bound uses the click amplitudes: since $I+U=2P$, the Hilbert–Schmidt triangle inequality gives $\sqrt{q}+\sqrt{r} \ge 2\sqrt{(sh)}$. Applying the same reasoning to no-click outcomes gives a second restriction.

Necessary and sufficient conditions

For $0<s<1$ and $0 \le q,r,h \le 1$, put $b=[(q+r)/2-sh]/(1-s)$ and $c=(q-r)/[4\sqrt{(s(1-s))}]$. The pair of click rates is feasible exactly when

> $0 \le b \le 1$, $c^2 \le hb$, $c^2 \le (1-h)(1-b)$.

To see necessity, purify ρ and split the purification into its normalized P and $I-P$ components. Their amplitudes are $\sqrt{s}$ and $\sqrt{(1-s)}$. Compress E tensored with identity on the purifying space to the two component vectors. This gives a two-dimensional effect with diagonals h and b. The real part of its off-diagonal is c because U changes the relative sign of the two components. Positivity of the effect and its complement gives the two determinant inequalities.

Conversely take the state with real amplitudes $\sqrt{s}$ and $\sqrt{(1-s)}$ and the real effect with diagonals h,b and off-diagonal c. The displayed inequalities make this a valid effect and direct multiplication reproduces q and r. The two-dimensional construction embeds into the joint probe space. If $s=0$ or 1, the reflection leaves the state unchanged, so $r=q$; at $s=1$ both equal h.

Why the upper endpoint matters

At fixed q the click determinant bounds r from below by $(2\sqrt{(sh)}-\sqrt{q})^2$ and from above by $(2\sqrt{(sh)}+\sqrt{q})^2$. Apply the same expressions to $1-r$, $1-q$ and $1-h$ for the no-click determinant. Intersect these ranges with [0,1] and the interval enforcing $0 \le b \le 1$. For the supplied inputs the active lower endpoint comes from the click determinant; the active upper endpoint comes from the no-click determinant.

At fixed q and h	Reflected rate	b	c
Lower endpoint	44.970404%	0.00195526	−0.04399668
Upper endpoint	55.102947%	0.08773340	−0.09551265

At the lower endpoint the effect has eigenvalues 0 and 0.99195526…. At the upper endpoint its complement is singular. The allowed section is an interval because the two-dimensional effect depends affinely on r and the effect set is convex. Averaging the two arms only would give the weaker lower endpoint 0.447394385938…. The sharp minimum increase over q is 0.086535922873…, or 8.65359229 percentage points.

The numerical interval is conditional on the same prepared state, a stable common effect, the specified q and h, and an independently characterized reflection about the joint match state. It does not predict an observed outcome for an uncharacterized local phase pulse. Statistical intervals or bounds on path-dependent errors require propagating those uncertainties through the feasible set. The original four-report table contains no reflected-arm observations.

Published hardware comparison

Aasen et al. found appreciable off-diagonal components in several noisy X- and Y-basis measurements on superconducting qubits. Most Z-basis reconstructions were consistent with statistical fluctuations in their off-diagonal components. Appendix E and Figure 12 identify the location in the preprint. These observations concern the authors’ reconstructed detector effects and do not characterize the sixteen-dimensional reflection considered here.

Aasen et al., Communications Physics 7, 301 (2024).

Preprint, Appendix E and Figure 12.